Credit Risk Portfolio Monitor: solution overview

A hosted dashboard for a risk analyst, built on 5 synthetic bank tables (14,685 loan accounts, 16.1% default rate), with an LLM assistant and anomaly detection embedded.

What the dashboard shows and how an analyst uses it

View	What it shows	How an analyst uses it
Portfolio	KPIs, insights, exposure and default by risk band, any dimension or two-way cross-cut, vintage and delinquency trends, riskiest accounts.	Filter to a segment, find where risk and money concentrate, select a row to drill into an account.
Account	Score, risk band, percentile, key drivers, 12-month repayment against peers, bureau lines, collections, data flags.	Understand why one account scores as it does before acting on it.
Ask the Portfolio	Plain-English questions answered with exact figures and a source. A guided query needs no LLM.	Get a figure without building a cut, and check how it was computed.
Anomaly Review	Ranked list of 313 accounts with reasons, peer comparison, decision form and log.	Work the list: read why, then Confirm or Dismiss with a note.
Model & Monitoring	AUC, KS, Gini, ROC, deciles, SHAP, sensitivity, PSI, fairness, monitoring plan.	Judge how far to trust the score, and spot drift.
Data Notes	Join logic, every data-quality decision with counts, assumptions.	Check how the numbers were made.

The two AI features

	Ask the Portfolio (Gemini)	Anomaly detection
How it works	Gemini only picks a query from a fixed catalogue, or declines. pandas computes the figures. Gemini then words the result, and code checks every number, replacing the wording if one is unsupported.	Six rule detectors and an Isolation Forest compare each account with its own history, its peers and its bureau record. The target is never used to fit them.
Trigger	An analyst submits a question.	Batch run. An account is listed when 2+ detectors agree: 313 accounts, 2.1%.
False positive	Answers what it should decline, or reads the question as a different metric, such as originated amount for balance.	A one-off payment slip, agreed hardship plan or data error. 67% of listed accounts did not default.
False negative	Declines an answerable question because the wording maps to no listed metric.	Steadily poor but normal for peers, or a clean-record defaulter. 96% of defaults are not listed: it does not replace the model.
Human step	The analyst checks the query and source under each answer and can mark it as wrong. Nothing changes as a result.	The analyst records Confirm or Dismiss with a note. Nothing reaches a customer without it. Dismiss rates tune thresholds.

Model results and approach

Model	AUC (95% interval)	Gini	KS	Out-of-time AUC
Logistic regression (main)	0.674 (0.661-0.687)	0.347	0.259	0.676
LightGBM (challenger)	0.671 (0.658-0.685)	0.343	0.251	0.678

24 features from customer and loan data, 12 months of repayment and date-limited bureau history, validated with 5 folds grouped by customer. LightGBM had to win by 0.005 AUC to replace the simpler model, and did not. The riskiest 10% of scores default at 36.4%, and the riskiest 30% hold 50% of defaults. Bureau data is cut off at the scoring date (ignoring dates would give AUC 0.676). Gender, marital status, region and age are excluded pending compliance review (age alone would add 0.004). The signal is modest by nature: a much higher AUC would suggest leakage. After deployment: monthly PSI on scores and features, quarterly AUC, KS, calibration and fairness checks by vintage, with set retraining triggers.

Top 3 portfolio insights

- **Mortgages are the concentration risk.** Mortgages are 9.8% of accounts but 52.3% of originated exposure and 65.0% of defaulted dollars. Their 24.7% default rate is the highest of any product (others: 13.9% to 16.2%) even though they are secured, and 240-month mortgages reach 29.2%. **So what:** Review mortgages first and tighten long-term underwriting. Collateral values are needed from the bank, because the actual loss cannot be sized without them.
- **Income and employment together separate risk.** Default falls from 32.4% under $40k to 9.5% at $100k+. Unemployed borrowers are 4.9% of accounts but 10.6% of defaults. Unemployed mortgages default at 54.3% (70 accounts, $15.6M). Contract borrowers run 35.1% under $40k but 12.8% at $60-80k. **So what:** Apply a hard policy rule to mortgages for unemployed applicants, and assess contract workers by income rather than treating them as one group.
- **Delinquency alerts miss nearly half of defaults, and collections are not targeted.** 46.7% of defaulters never missed a payment in 12 months. Only 38.0% of defaulted accounts had any collections action, while 3,380 non-defaulted accounts did. Recoveries equal 0.99% of defaulted loan amounts. **So what:** Prioritise collections by model score rather than by missed payments alone, and audit the collections data first: 44.9% of events are dated before the loan began.

Key assumptions

The target has no date, so default is assumed to come after month 12, making this a month-12 behaviour score (3- and 6-month sensitivity is reported). The data is assumed to be a 2025-12-31 snapshot. Status and balances on active bureau lines have no reporting date. Latest balance is approximate, because it falls by a fixed 60% of each payment. Free hosting can reset the decision log on restart.

What production would need from the bank's IT and data teams

- **Data:** a dated default flag and definition (such as 90+ days past due), collateral values and recoveries to size loss, bureau data with reporting dates, and calendar dates on repayments. Collections data needs fixing: 45% of events pre-date their loan.
- **Platform:** automated feeds, a governed database for decisions with analyst sign-in and an audit trail, access control, a model registry and secrets management.
- **LLM and governance:** an approved enterprise LLM provider where content is not used for training, with data-residency review and spend limits. Model validation, and compliance review of fair-lending choices, especially age.